

Human Approval as a Controllable Simulation Mechanism

SocioVerse Challenge 2026 — Track C: Methodological and Systems Innovation

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Status: Research proposal; experiments and results have not yet been produced.

Question and contribution

Question: In a multi-agent environment, when do explicit human-review checkpoints prevent unsafe execution, and when do they create delay, reviewer overload or unfair allocation of attention? Many simulated agent workflows assume approvals are instantaneous or model only the final outcome. We propose making review capacity, information quality, escalation and authority boundaries explicit variables in a reproducible SocioVerse2 study.

Our intended contribution is a **configurable approval-policy adapter** with transparent state transitions and metrics. This is motivated by existing BOLT-controlled workflow prototypes, but the SocioVerse2 study, integration and validation are **future work**, not deployed features or existing experimental findings.

Population, environment and behavior (P/E/B)

Population (P). 60–120 synthetic requesters, operators and reviewers. Requesters vary in priority and risk; reviewers vary in expertise, review capacity, tolerance for uncertainty and consistency. All agent records are synthetic and generated with documented random seeds.

Environment (E). Discrete time steps; task queues, deadlines, constrained review capacity, severity of consequences and incomplete supporting evidence. Every experimental condition receives the same input event sequence and resource budget where comparisons require it.

Behavior (B). Requesters propose an action with supporting evidence; operators queue or execute; reviewers approve, request clarification, reject or escalate. Approval **requested** and approval **granted** are separate events. The simulator records reasons, trace IDs, time spent and subsequent outcomes.

Interventions. Three arms: A, ungated execution; B, static deterministic risk rules; C, a capacity-limited, synthetic human-review proxy with explicit escalation. We will vary incoming workload, reviewer capacity, evidence completeness, risk threshold and agent heterogeneity. The simulated reviewer is **not an actual human participant**.

Evaluation design and data

We plan at least five fixed seeds (11, 29, 47, 83, 101), identical workloads for matched comparisons, and robustness tests across parameter ranges. Baselines include ungated execution, rule-only gating and a FIFO review queue. Ablations will remove escalation, reviewer heterogeneity and evidence traceability one at a time.

Metrics: synthetic unsafe-action rate under a declared oracle; throughput; median and 95th-percentile latency; reviewer utilization; false denials; review coverage; explanation completeness; and performance differences between synthetic priority groups. We will compare benefits against delays and capacity cost, and report uncertainty instead of claiming an advantage before running experiments.

Data sources: reproducible synthetic populations and event generators only; public research on bounded rationality, queues and organizational accountability will inform assumptions and be cited in the final paper. No real individuals, customer logs, private records or unlicensed datasets. We will not interpret synthetic effects as real-world causal evidence.

Execution, milestones and reproducibility

We plan to build the mandatory `studies/unfire_approval_gates/` layout with `study.yaml`, `population/`, `environment/`, `simulation/`, `trajectory/`, `reports/`, `run.py` and `README.md`. First deliver a deterministic offline `llm_kind: scripted` run with no paid API key. Pin dependencies and seeds, publish generator source and CSV trajectories, add tests for state transitions and illegal approvals, and provide a one-command reproduction path. If an LLM is later used, disclose model, calls, prompts and estimated cost. All contest study code and derived data will be released under an open-source license.

9–12 Oct: study scaffold and deterministic smoke test. **13–21 Oct:** treatment arms, baselines, quantitative tests. **22–29 Oct:** sensitivity checks, 4–8 page report and real demonstration (maximum 5 minutes). **By 31 Oct:** submit all final materials.

Limits: no real human judgments are simulated faithfully; risk oracles may bias results; reviewer behavior is simplified; conclusions will be bounded by assumptions and validated only within the declared synthetic testbed.

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